

DFI command encode — full command-bus truth table (JESD79-5 Table 241)

One channel, DDR5, 1:1 gear, N_RANKS=2, target = rank0. CA0–CA13 = the shared CA opcode/address bus; CSn/CKE/ODT/PAR/2N/RST = command sideband. green =fixed opcode bit; yellow =field / conditional; blue =static (one value, no per-phase mux). H=1, L=0 (active-low asserted), V=valid, X=don't care. CLK=beat (0=first CK, 1=second; CA is SDR so 1 CK = 1 command beat).

1 2-CLK commands (CA1 = L; two CK beats)

cmd	CLK	CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	CSn0	CSn1	CKE0	CKE1	ODT0	ODT1	PAR	2N	RST
Activate ACT	0	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
	1	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	R17	H	H	H	H	L	L	P	cfg	H
Write WR	0	H	L	H	H	L	BL	BA0	BA1	BG0	BG1	BG2	CID	CID	CID	L	H	H	H	w	w	P	cfg	H
	1	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP	WRp	V	CID	H	L	H	H	w	w	P	cfg	H
Read RD	0	H	L	H	H	H	BL	BA0	BA1	BG0	BG1	BG2	CID	CID	CID	L	H	H	H	w	w	P	cfg	H
	1	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	AP	V	V	CID	H	L	H	H	w	w	P	cfg	H
MR Write MRW	0	H	L	H	L	L	MR0	MR1	MR2	MR3	MR4	MR5	MR6	MR7	V	L	H	H	H	L	L	P	cfg	H
	1	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	H	H	H	H	L	L	P	cfg	H
MR Read MRR	0	H	L	H	L	H	MR0	MR1	MR2	MR3	MR4	MR5	MR6	MR7	V	L	H	H	H	w	w	P	cfg	H
	1	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	H	L	H	H	w	w	P	cfg	H

CSn (one-cold, active-low): ck0 selects target rank (rank0=L, rank1=H). ck1: ACT/MRW =H (no-ODT 2-clk, note 11); WR/RD/MRR ck1 CSn on the non-target rank = L to assert its ODT (note 8). ODT w = asserted in the WL-aligned write window (ODT_GEN), not fixed to the command beat. PAR P = even parity over CA0..CA13 that beat. WR vs RD differ only at CA4 (rw picks it); ap=CA10 ck1.

2 1-CLK commands (CA1 = H, CA0 = H; one CK beat)

cmd	CLK	CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12	CA13	CSn0	CSn1	CKE0	CKE1	ODT0	ODT1	PAR	2N	RST
Refresh all	0	H	H	L	L	H	CID3	V	V	RIR	H	L	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
REFab																								
RFM all RFMab	0	H	H	L	L	H	CID3	V	V	V	L	L	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
Refresh sb	0	H	H	L	L	H	CID3	BA0	BA1	RIR	H	H	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
REFsb																								
RFM sb RFMsb	0	H	H	L	L	H	CID3	BA0	BA1	V	L	H	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
Precharge all	0	H	H	L	H	L	CID3	V	V	V	V	L	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
PREab																								
Precharge sb	0	H	H	L	H	L	CID3	BA0	BA1	V	V	H	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
PREsb																								
Precharge pb	0	H	H	L	H	H	CID3	BA0	BA1	BG0	BG1	BG2	CID	CID	CID	L	H	H	H	L	L	P	cfg	H
PREpb																								
Self-Refresh	0	H	H	H	L	H	V	V	V	V	H	L	V	V	V	L	H	L	H	L	L	P	cfg	H
entry SRE																								
SR entry freq	0	H	H	H	L	H	V	V	V	V	L	L	V	V	V	L	H	L	H	L	L	P	cfg	H
SREF																								
Power-Down	0	H	H	H	L	H	V	V	V	V	V	H	ODT	V	V	L	H	L	H	H	L	P	cfg	H
entry PDE																								
MPC	0	H	H	H	H	L	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	L	H	H	H	L	L	P	cfg	H
NOP	0	H	H	H	H	H	V	V	V	V	V	V	V	V	V	L	H	H	H	L	L	P	cfg	H
Power-Down exit	0	H	H	H	H	H	V	V	V	V	V	V	V	V	V	L	H	H	H	L	L	P	cfg	H
PDX																								
Deselect DES	0	CA = X (deselect)														H	H	H	H	L	L	P	cfg	H

REF/RFM: CA0–CA4 = 11001; CA9=REF(H)/RFM(L), CA10=sb(H)/ab(L). PRE: CA0–CA3=1101; CA4=pb(H)/sb-ab(L), CA10=sb(H)/ab(L). CKE0=L for SRE/SREF/PDE (rank0 enters low-power; PDX drives it back H). PDE ODT0=H = ODT persists in power-down (note 16). NOP has CSn=L; DES = CSn all H (nobody decodes CA). Static 2N/RST = one value every phase, boot/init driven.

DFI command sideband ports — CS_n / CKE / ODT / parity / static

One channel, DDR5. Everything on the command bus that is *not* the CA opcode (see `opcode_rom.pdf` for the CA truth table). green =fixed level, yellow =conditional. L=0 (active-low asserted), H=1, -=no 2nd beat. **ck0**=first CK beat, **ck1**=second (2-clk commands only; 1 CK = 1 command beat, CA is SDR).

1 dfi_cs_n_pN[N_RANKS-1:0] — rank select, ONE-COLD (active-low)

The **rank** field of the phaser record decodes to a one-cold vector: the target rank's bit = L, every other rank = H. `dfi_cs_n[r] = !(r == rank)` on the select beat. CS_n is the *qualifier* — the DRAM only decodes the CA bus on the CK beat its CS_n is L. Below = the **target rank bit** per command (non-target ranks = H, except the ck1-ODT case).

command	CS_n ck0	CS_n ck1	note
ACT / MRW / WRP (2-clk, no ODT)	L	H	2-clk with no ODT control: CS_n <i>must</i> be H on ck1, else DRAM ignores the command (JEDEC note 11).
RD / WR / MRR (2-clk, ODT)	L	L / H	ck1 CS_n controls ODT on <i>non-target</i> ranks (note 8): drive L on a non-target rank to assert its ODT during the data window; H = no ODT.
PRE(ab/sb/pb), REF(ab/sb), RFM(ab/sb)	L	-	1-clk command: single select beat.
MPC / SRE / SREF / PDE / PDX / NOP	L	-	1-clk command.
DES (deselect)	H	H	all ranks H = nobody decodes the CA bus this beat.

One-cold vs one-hot: CS_n is one-cold (exactly one bit **low**). A binary `rank[clog2 N_RANKS]` → NR-wide decoder → invert → one-cold. Two different-rank commands in one bundle each drive their own rank bit low on their own phase, so a rank-0 CAS + rank-1 ACT coexist (this is the cross-rank parallelism). CKE and ODT are *not* one-hot — they are per-rank independent state (any subset may be asserted).

2 Per-rank state ports (POWER_FSM / ODT_GEN — not opcode-driven)

These do not follow the command opcode; they follow power state or the write data window. Each is a per-rank vector, driven per phase.

port	driver	value / when asserted
dfi_cke_pN[NR-1:0]	POWER_FSM	H = clock enabled (normal). Driven L for a rank on Power-Down entry (PDE) / Self-Refresh entry (SRE); held L until PDX/SRX. Per-rank independent, latency-critical (CKE setup).
dfi_dram_clk_disable_pN[NR-1:0]	POWER_FSM	H = gate CK to that rank (deep power-down / self-refresh, after CKE low). Per rank.
dfi_odt_pN[NR-1:0]	ODT_GEN	Asserted in a WL-aligned window around a WRITE (offset = odt_latency). Target and/or non-target ranks per RTT_WR/RTT_NOM/RTT_PARK. Timing-driven, not command-bit driven; overlaps the CAS by CWL.

3 Computed / static command ports

port	kind	value
dfi_parity_pN	per phase	CA-bus parity for that beat: = ~(CA0..CA13) (even parity) when parity_en ; else 0. One bit per phase, recomputed from the CA vector the encoder just built.
dfi_2n_mode	static	1 = 2N command timing (CA held 2 CK, CS pulses the sample beat — SI margin, half command rate). Config bit, boot/init only. One value all phases, no per-phase mux.
dfi_reset_n	static	L during DRAM reset, H after; driven by INIT_FSM, one-way. Not phased.

4 How each maps onto the phases (gear)

Same rule as the CA bus. Each *phased* port has one mux per phase; **phase_sel** routes the right {slot, ck-beat} value onto phase pN. At **1:1 (gear=00)** there is one phase, so a 2-clk command's ck0/ck1 land in two successive mc_clks (temporal, held by **two_clk/repeat**). At **1:2 / 1:4** the beats land on parallel phases in one mc_clk (constant spatial wiring: p0=ck0, p1=ck1, ...). **Static** ports (**2n_mode**, **reset_n**) drive one value across all phases —

no mux. A **1-clk** command uses one phase; the next phase is free for another command.